Theorem 1. *Let R be a finite-dimensional (possibly noncommutative) $k[\varepsilon]$-algebra such that $\operatorname{ann}(\varepsilon)^2 = 0$ and $S := R/\operatorname{ann}(\varepsilon)$ is left self-injective. Then R admits a finite-dimensional faithful left module M which is free as a $k[\varepsilon]/(\varepsilon^2)$-module.*

Proof. The data of an associative $k[\varepsilon]$-algebra R with $\operatorname{ann}(\varepsilon)^2 = 0$ is equivalent to the data of the associative k-algebra $S := R/\operatorname{ann}(\varepsilon)$, the S-bimodule $I := \operatorname{ann}(\varepsilon)$, a square-zero extension $0 \to I \to R \to S \to 0$, and an injective S-bimodule map $u : S \hookrightarrow I$ sending 1 to ε. Therefore it suffices to show: for self-injective finite-dimensional S with an injection $u : S \hookrightarrow I$ of S-bimodules and a square-zero extension of S by I, there is a finite-dimensional faithful left R-module free over $k[\varepsilon]/(\varepsilon^2)$.

Let $S^e := S \otimes_k S^{\mathrm{op}}$. By embedding I into an injective S^e-module and replacing R by the corresponding pushout extension, we may assume I is injective as an S^e-module; restriction of scalars reduces the general case to this one. Then the square-zero extension splits: since square-zero extensions are classified by derivations $S \to I[1]$, the obstruction group $\operatorname{Ext}^1_{S^e}(\mathbb{L}_S^{E_1}, I) \cong \operatorname{Ext}^2_{S^e}(S, I)$ vanishes for injective I. Hence $R \cong S \ltimes I$.

Now take $V := S^{\oplus m}$ for some m, so that $\operatorname{End}_k(V) \cong \operatorname{End}_k(S)^m$ as S^e-modules. Since S is finite-dimensional self-injective, $\operatorname{End}_k(S)$ is an injective cogenerator for S^e-modules; enlarging m if needed, choose an S^e-monomorphism $\beta : I \hookrightarrow \operatorname{End}_k(V)$ extending the left-action map along u (equivalently $\beta(u(1)) = \mathrm{id}_V$).

Set $M := k[\varepsilon]/(\varepsilon^2) \otimes_k V$. Let S act on M through its action on V, and let $i \in I$ act on M by multiplication by ε followed by $\beta(i)$ on V. Then $u(1) \in R$ acts as scalar multiplication by ε on M, so M is free over $k[\varepsilon]/(\varepsilon^2)$. If $(s, i) \in S \ltimes I$ acts by 0 on M, then reducing modulo ε gives $s = 0$, and then $\beta(i) = 0$ forces $i = 0$; thus M is faithful. □